

Multidimensional routine hematologic modeling for acute myocardial infarction phenotype discrimination in coronary artery disease: a single-center retrospective study

Jibulang Ainiwaer¹, Mubareke Pidamaimati¹, Qian Xie¹, Qixing Pi¹, Wei Ji¹, Binbin Fang¹, Fen Liu¹, Long Zhao¹, Xiaomei Li^{1,2*}, Yining Yang^{3,4*}

1 Department of Cardiology, The First Affiliated Hospital of Xinjiang Medical University, Urumqi 830054, Xinjiang, China.

2 Xinjiang Key Laboratory of Cardiovascular Disease Research, Clinical Medical Research Institute of First Affiliated Hospital of Xinjiang Medical University, Urumqi 830054, China.

3 Department of Cardiology, People's Hospital of Xinjiang Uygur Autonomous Region, Urumqi 830001, Xinjiang, China.

4 Xinjiang Key Laboratory of Cardiovascular Homeostasis and Regeneration Research, Urumqi, Xinjiang, China.

Corresponding authors: Xiaomei Li (primary corresponding author), lixm505@163.com; Yining Yang (co-corresponding author), yangyn5126@xjrmyy.com.

Abstract

Background: Acute myocardial infarction (AMI) is accompanied by inflammatory, platelet, coagulation, and erythrocyte responses. Routine complete blood count (CBC) measurements capture several of these changes but are often reduced to fixed ratios. We examined whether joint modeling of routine hematologic measurements could improve AMI phenotype discrimination beyond the pan-immune-inflammation value (PIV) in patients with coronary artery disease.

Methods: This single-center retrospective study included 1,820 patients with a discharge-diagnosis-defined AMI or non-AMI CAD phenotype. Core-7 jointly modeled seven routine CBC variables using elastic-net logistic regression. Internal performance was evaluated by stratified

outer five-fold cross-validation repeated 10 times, with five-fold tuning within each training set. PIV was the prespecified primary conventional comparator; fibrinogen was assessed in an Enhanced model. All reported internal estimates used patient-level mean held-out probabilities. Results: The primary cohort comprised 1,820 patients (453 AMI and 1,367 non-AMI CAD). PIV had an AUC of 0.708 (95% CI 0.678-0.735); Core-7 had an AUC of 0.725 (0.697-0.753), a modest paired increase of 0.017 (0.003-0.031). The full-cohort Enhanced model had an AUC of 0.735 (0.710-0.763). Conclusions: Joint modeling of routine hematologic measurements provided modest additional information for AMI phenotype discrimination beyond PIV in this single-center retrospective cohort. Fibrinogen contributed a further small increment. These findings do not establish diagnostic or triage readiness; independent cohorts with prospectively time-anchored laboratory measurements and standardized clinical adjudication are required. Keywords: acute myocardial infarction; coronary artery disease; complete blood count; hematologic biomarkers; fibrinogen; internal validation; calibration

Background

Acute myocardial infarction (AMI) in patients with coronary artery disease (CAD) reflects the interaction of plaque inflammation, thrombosis, platelet activation, myocardial injury, and systemic stress [1,2]. These processes are accompanied by coordinated changes in circulating blood cells and coagulation proteins. Neutrophilia and lymphocyte depletion may accompany acute inflammatory and neurohumoral stress responses, monocytes participate in plaque inflammation and subsequent immune activation, and platelet turnover and activation can alter both platelet count and mean platelet volume (MPV) [1,3-14]. Erythrocyte indices such as red cell distribution width (RDW) and hemoglobin may reflect erythropoietic stress, systemic inflammation, comorbidity, and oxygen-delivery reserve, while fibrinogen links the acute-phase response with thrombosis [15-17]. Because these complete blood count (CBC) and

coagulation measurements are obtained routinely in cardiovascular care, they provide a clinically accessible window into several biological processes relevant to AMI. Previous studies have mainly evaluated hematologic inflammation through derived indices such as the neutrophil-to-lymphocyte ratio (NLR), platelet-to-lymphocyte ratio (PLR), systemic immune-inflammation index (SII), systemic inflammation response index (SIRI), and pan-immune-inflammation value (PIV) [3-12]. These indices are inexpensive and easy to calculate, but each applies a fixed mathematical weighting to a limited set of cell counts. Other routinely available measurements, including MPV, RDW, and hemoglobin, are therefore often considered separately even though they may provide complementary information about the acute coronary state [13-16].

We hypothesized that evaluating these measurements together would better represent the multidimensional hematologic response accompanying AMI than a single predefined ratio. The Core-7 model was therefore built around three clinically recognizable components: leukocyte-related inflammatory responses (neutrophils, lymphocytes, and monocytes), platelet-related features (platelet count and MPV), and erythrocyte-related features (RDW and hemoglobin). Fibrinogen was examined separately as an additional coagulation and acute-phase marker [13-17]. This biologically informed grouping provided the rationale for joint modeling rather than simply selecting variables from the available laboratory database.

Accordingly, we evaluated a multidimensional routine hematologic model for discrimination of an AMI phenotype among patients with CAD. We compared the CBC-only Core-7 model with prespecified conventional indices, particularly PIV, and assessed the additional information associated with fibrinogen. All analyses were phenotype-discrimination analyses rather than future-event prediction.

Methods

Study design and setting

This was a single-center retrospective observational study at the Department of Cardiology, The First Affiliated Hospital of Xinjiang Medical University, Urumqi, China. The source dataset was described as comprising patients undergoing coronary angiography from 1 January 2020 through 1 January 2026. The extraction query and complete screening ledger were not recovered, so the stated period and eligibility could not be independently machine-verified and consecutive enrollment is not claimed. The research extract contained deidentified dates; selected laboratory measurements could not be uniformly linked to the index angiography hospitalization.

Eligibility and study population

The source export contained 2,548 records for 2,279 unique patients. Of these, 2,010 patients had one source record and 269 had two source records. One record per patient was retained using a completeness-based selection rule rather than an earliest- or admission-first rule. Discharge-diagnosis text was available for 1,944 patients; 456 met the definite AMI text phenotype, 1,372 met the definite non-AMI CAD phenotype, 5 had an ambiguous MI mention, and 111 had no qualifying CAD/angina/MI phenotype. A further 335 lacked a discharge diagnosis. The required absolute neutrophil, lymphocyte, monocyte, and platelet measurements were available for 453 of 456 AMI and 1,367 of 1,372 non-AMI CAD patients, yielding the primary cohort of 1,820.

Outcome definition

AMI phenotype was derived from discharge-diagnosis text using prespecified rules. Definite AMI required explicit current acute or subacute MI wording, including unambiguous STEMI, NSTEMI, or an acute wall-specific description. Old/prior MI, history of MI, prior PCI, CAD,

stable angina, or unstable angina alone did not define AMI. MI mentions without clear acute or old/prior context and uncertainty wording were not classified as definite AMI. No independent cardiologist, ECG, imaging, or uniform troponin adjudication was available for the full cohort; the phenotype therefore represents a discharge-diagnosis-text classification, not formal adjudication under the Fourth Universal Definition [2].

Primary and strict-control cohorts

The primary cohort required non-missing absolute neutrophil, lymphocyte, monocyte, and platelet counts. The strict/high-specificity sensitivity comparison included 138 later-group AMI records meeting the prespecified discharge-text rule and 288 strict CAD/angina controls without current MI, old/prior MI, or an ACS-only marker (N=426). The stricter AMI rule required an MI keyword plus an acute marker, excluded subacute and uncertain wording, and required a STEMI/NSTEMI/ST-segment subtype or explicit acute-MI phrase/site. These criteria were based on discharge text and use information available at or after discharge; they are not a prospective baseline rule or independent clinical adjudication.

Predictors and laboratory measurements

The Core-7 predictors were absolute neutrophil, lymphocyte, and monocyte counts, platelet count, MPV, RDW, and hemoglobin; fibrinogen was added only for the Enhanced model. Recorded units were $10^9/L$ for cell counts, fL for MPV, percent for RDW, and g/L for hemoglobin and fibrinogen. For patients with multiple source records, one patient-level record was selected by prioritizing a non-empty discharge diagnosis, greater completeness of absolute neutrophil, lymphocyte, monocyte, and platelet measurements, availability of a parsable WBC/CBC timestamp, higher overall non-missingness, and original source-row order as the final tie-break. The extract did not provide sufficient encounter-level identifiers or angiography/diagnosis timestamps to determine uniformly whether a selected measurement was the first admission sample or preceded angiography, diagnosis, or treatment. All 1,820

selected measurements therefore remain timing-unverified relative to the index angiography hospitalization and are interpreted as patient-level routine laboratory features associated with the discharge-diagnosis AMI phenotype.

Conventional inflammatory indices

Using absolute counts, we calculated $NLR = \text{Neut}/\text{Lymph}$, $PLR = \text{PLT}/\text{Lymph}$, $MLR = \text{Mono}/\text{Lymph}$, $SII = \text{PLT} \times \text{Neut}/\text{Lymph}$, $SIRI = \text{Neut} \times \text{Mono}/\text{Lymph}$, $PIV = \text{PLT} \times \text{Neut} \times \text{Mono}/\text{Lymph}$, and $HRR = \text{Hb}/\text{RDW}$. PIV was the prespecified primary conventional comparator; its prespecified $\log(1+PIV)$ transformation and training-fold standardization were applied in an unpenalized logistic model on the same primary cohort and outer-fold partitions as Core-7. All indices remained continuous and were not dichotomized at data-derived cutoffs [3-17].

Model development and preprocessing

Core-7 used the seven CBC predictors, and the Enhanced model additionally included fibrinogen. Elastic-net logistic regression evaluated inverse-penalty C values of 0.10, 1.00, and 10.00 and mixing parameters of 0.25, 0.50, and 0.75 with tuning performed by five-fold cross-validation within each outer training set. Neutrophils, lymphocytes, monocytes, platelets, and fibrinogen were $\log(1+x)$ transformed and standardized; MPV, RDW, and hemoglobin were standardized on their original scale. Core-7 was complete for all 1,820 primary-cohort patients. In the full-cohort Enhanced analysis, missing fibrinogen was imputed using a median estimated within each training fold. A paired complete-case Core/Enhanced comparison was also conducted in the same 1,705 patients [18,19].

Internal validation and uncertainty

Primary internal validation used stratified outer five-fold cross-validation repeated 10 times, with five-fold tuning confined to each outer training set. Each patient received one held-out

probability per repeat; the final patient-level out-of-fold probability was the arithmetic mean of the 10 held-out probabilities. Identical outer partitions were used for paired model comparisons where applicable. Imputation, transformations, centering/scaling, and hyperparameter selection were estimated in training folds only. AUC, Brier score, calibration intercept, and calibration slope were calculated from the mean out-of-fold predictions. Confidence intervals were obtained from 1,000 patient-level bootstrap resamples of the fixed mean predictions and outcomes, without refitting the full model-development pipeline; they therefore do not include all model-development uncertainty. No univariable P-value screening or stepwise selection was used [20-26].

Temporal sensitivity analysis

An exploratory deidentified date-axis sensitivity analysis used the prespecified development, buffer, and later group allocation: development N=1,001 (AMI 196), buffer N=271 (AMI 79), and later N=548 (AMI 178). The earlier group was used for model development and the later group was evaluated as a fixed held-out date-axis group in the existing analysis. Because available dates could not be uniformly linked to the index angiography hospitalization, this analysis is not formal temporal validation and does not establish transportability.

Phenotype and missingness sensitivity analyses

High-specificity phenotype and strict-control results were treated as sensitivity analyses. The high-specificity comparison comprised 138 of the 178 later-group AMI records plus 288 strict CAD/angina controls; the other 40 later-group AMI records did not meet the prespecified diagnosis-text rule and should not be interpreted as adjudicated misclassifications. Fibrinogen availability and date linkage were audited; no troponin model was fitted because assay platform, ULN harmonization, serial change, and index timing were not sufficiently standardized.

Reporting and reproducibility

An exploratory decision-curve analysis was added using the same fibrinogen-complete N=1,705 patients and mean out-of-fold predictions for Clinical, Clinical+PIV, Clinical+Core-7, and Clinical+Enhanced models, with treat-all and treat-none references over thresholds 0.05-0.50 [27]. No threshold was optimized and no clinical utility or treatment benefit is inferred. Figures were generated in R from approved aggregate analysis outputs, each with independent source data and a reproducible script [31,32]. Reporting followed STROBE, with TRIPOD+AI and PROBAST+AI used as reporting and self-audit guidance [28-30]. Detailed definitions, calibration groups, predictor stability, comparator results, and sensitivity analyses are provided in Additional files 1 and 2.

Use of generative AI tools

Generative AI tools were used to assist with English-language polishing and code drafting under author supervision. All analysis code was reviewed and tested by the authors, and all data handling, statistical analyses, numerical results, references, interpretations, and final manuscript content were verified by the human authors, who retain full responsibility for the work. No generative AI was used to create patient data or scientific figures.

Results

Study population

The source export contained 2,548 records and 2,279 unique patients. A total of 2,010 patients had one source record and 269 had two; one patient-level record was retained using the completeness-based rule. Among 1,944 patients with a discharge diagnosis, 456 met the definite AMI text definition, 1,372 met the definite non-AMI CAD text definition, 5 had an ambiguous MI mention, and 111 did not meet the CAD/AMI phenotype target; 335 additional patients lacked a discharge diagnosis. Eight patients with a definite AMI or non-AMI CAD

phenotype lacked one or more required core CBC measurements (3 AMI and 5 non-AMI CAD), leaving 1,820 patients (453 AMI and 1,367 non-AMI CAD). The strict/high-specificity comparison contained 426 patients (138 AMI and 288 strict controls). Figure 1 summarizes record consolidation and the primary cohort.

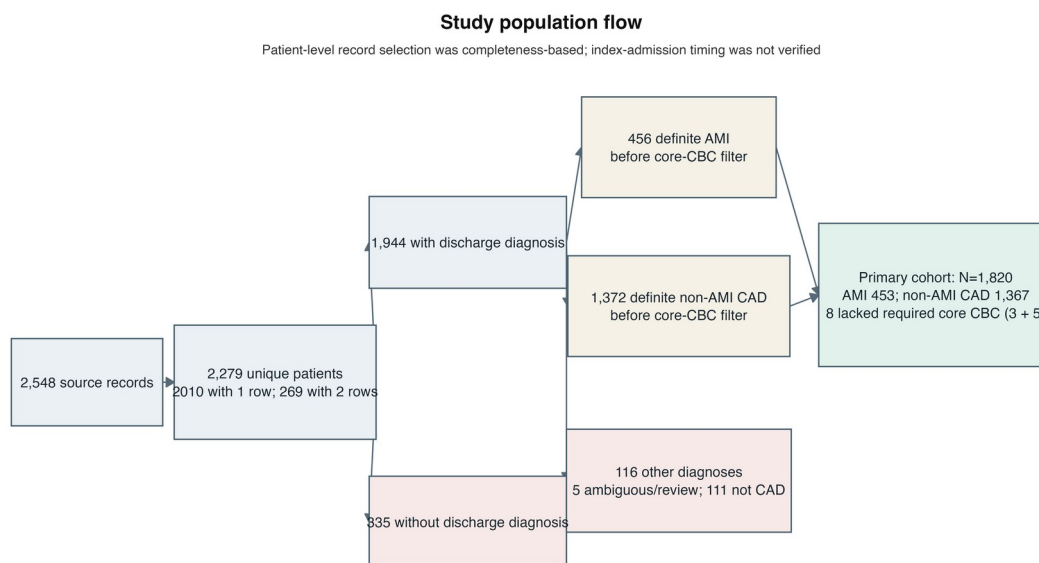


Figure 1. Study population flow and completeness-based patient-level record consolidation.

Table 1. Baseline characteristics of the primary cohort according to AMI phenotype

Variable	Unit	Overall (N=1820)	Definite AMI (N=453)	Definite non-AMI CAD (N=1367)	P-value
Age	years	58.8 [50.4, 67.7]	57.4 [48.9, 66.6]	59.3 [50.9, 68.2]	0.004
Male sex	n/N (%)	1452/1820 (79.8%)	376/453 (83.0%)	1076/1367 (78.7%)	0.049
Hypertension	n/N (%)	927/1741 (53.2%)	201/435 (46.2%)	726/1306 (55.6%)	<0.001
Diabetes	n/N (%)	459/1772 (25.9%)	109/442 (24.7%)	350/1330 (26.3%)	0.491
Neutrophils	10 ⁹ /L	4.4 [3.5, 6.0]	5.7 [4.0, 8.5]	4.2 [3.3, 5.3]	<0.001
Lymphocytes	10 ⁹ /L	2.0 [1.5, 2.5]	1.8 [1.3, 2.4]	2.0 [1.6, 2.5]	<0.001
Monocytes	10 ⁹ /L	0.5 [0.4, 0.7]	0.6 [0.4, 0.8]	0.5 [0.4, 0.6]	<0.001

Variable	Unit	Overall (N=1820)	Definite AMI (N=453)	Definite non-AMI CAD (N=1367)	P-value
Platelets	10 ⁹ /L	215.0 [178.0, 255.0]	224.0 [183.0, 266.0]	212.0 [175.5, 253.0]	<0.001
MPV	fL	10.3 [9.5, 11.1]	10.3 [9.6, 11.1]	10.3 [9.5, 11.1]	0.673
RDW	%	13.1 [12.6, 13.6]	12.9 [12.4, 13.4]	13.1 [12.7, 13.6]	<0.001
Hemoglobin	g/L	142.0 [132.0, 152.0]	144.0 [134.0, 155.0]	141.0 [131.0, 152.0]	0.009
Fibrinogen	g/L	3.5 [3.1, 4.1]	3.8 [3.2, 4.5]	3.5 [3.1, 3.9]	<0.001

Continuous variables are presented as median [interquartile range]. Age was available for all 1,820 patients and was derived using the primary birth date (baseline birth-date fallback), with admission date as reference when available and CBC/WBC timestamp otherwise; age was elapsed days divided by 365.2425, restricted to 18-120 years. P-values comparing definite AMI with definite non-AMI CAD were calculated using the Mann-Whitney U test for continuous variables and Pearson's chi-square test for categorical variables, with Fisher's exact test used for sparse expected cell counts. Available-case denominators for hypertension, diabetes, and fibrinogen were 1,741, 1,772, and 1,705, respectively. These P-values were descriptive and were not used for predictor selection or model development.

Hematologic model

Core-7 jointly modeled immune-inflammatory, platelet-related, and erythrocyte-related measurements; fibrinogen was added as a secondary coagulation component in the Enhanced model. These groupings organize clinical interpretation and are not treated as independent causal pathways.

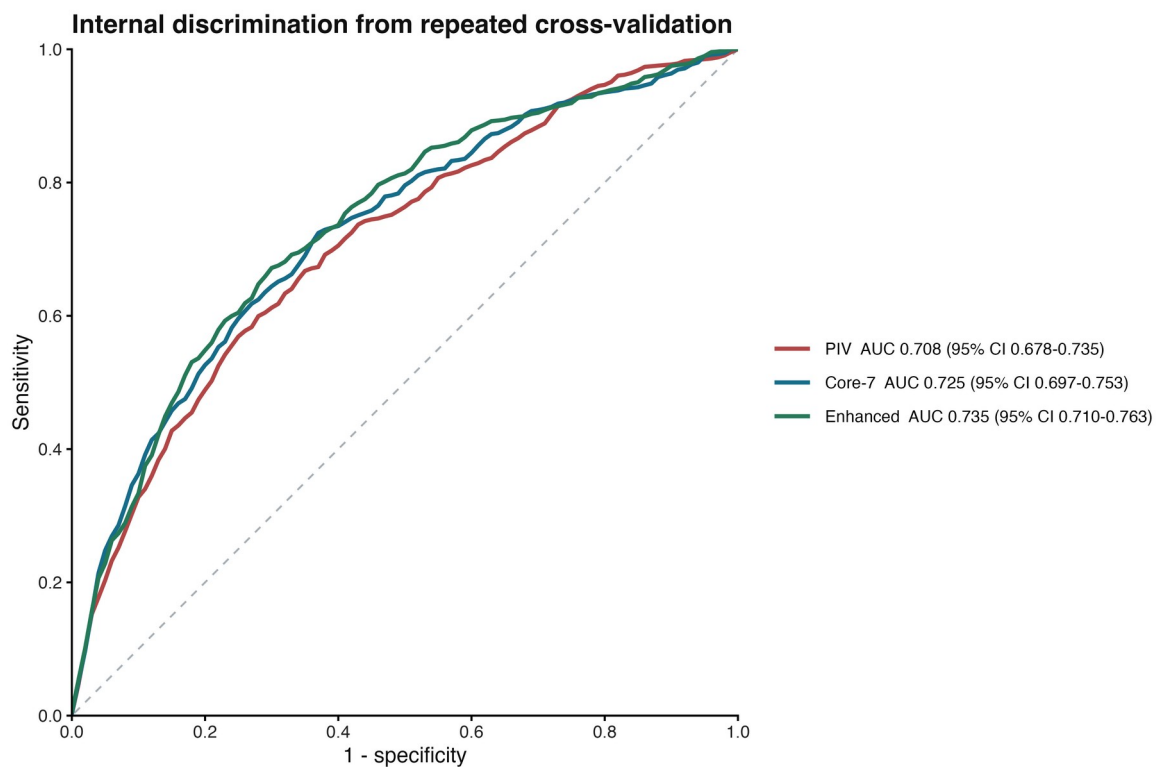
Internal model performance

Among the prespecified conventional indices, PIV had the highest AUC (0.708, 95% CI 0.678-0.735). Core-7 had an AUC of 0.725 (0.697-0.753); its paired increase over PIV was modest (Δ AUC 0.017, 95% CI 0.003-0.031). The full-cohort Enhanced model, with training-fold fibrinogen imputation, had an AUC of 0.735 (0.710-0.763); its paired increment over Core-7 was 0.010 (95% CI 0.003-0.017). Core-7 Brier score was 0.163, calibration intercept -0.026, and calibration slope 0.974. Figure 2 shows ROC curves and Figure 3 shows calibration from the same mean 5-fold-by-10-repeat held-out predictions.

Table 2. Internal cross-validated model performance and paired comparisons

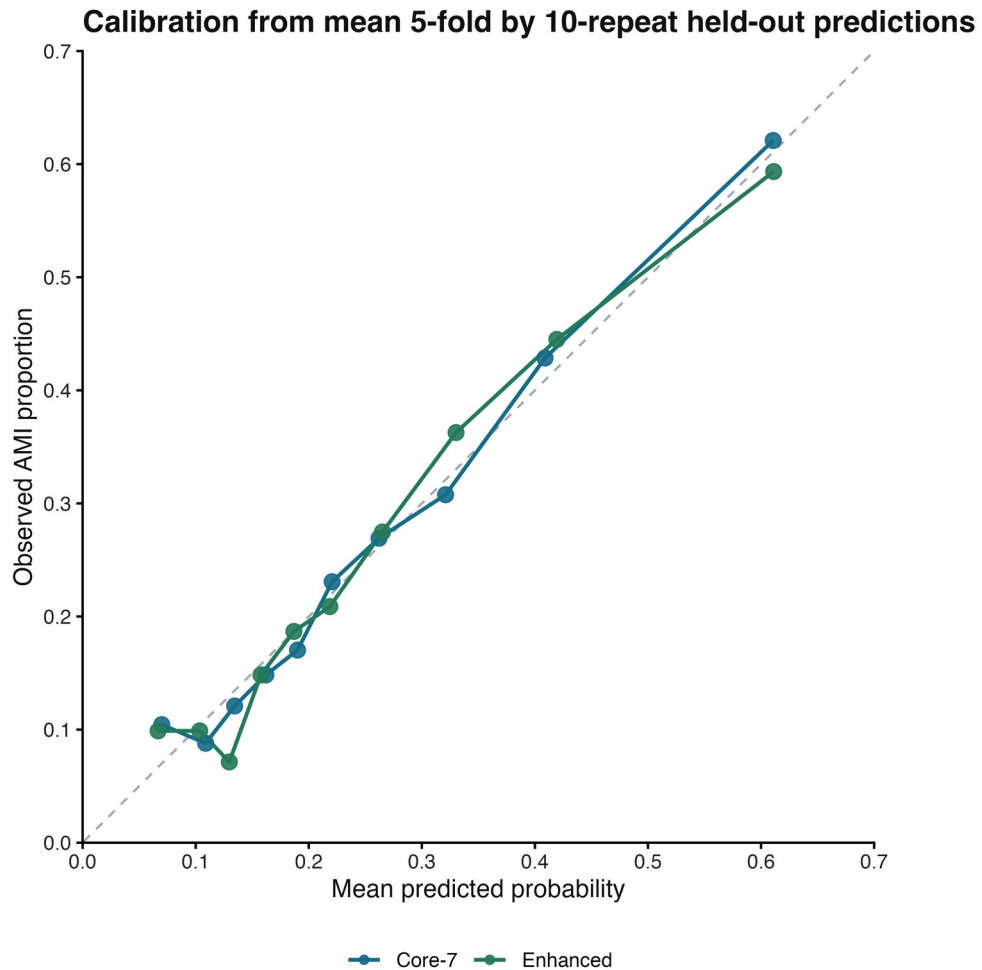
Model / comparis on	N	AMI	AUC (95% CI)	Δ AUC (95% CI)	Brier	Calibratio n intercept	Calibratio n slope
PIV	1820	453	0.708 (0.678- 0.735)	—	0.167	-0.008	0.992
Core-7	1820	453	0.725 (0.697- 0.753)	—	0.163	-0.026	0.974
Enhanced (full cohort; training- fold fibrinogen imputatio n)	1820	453	0.735 (0.710- 0.763)	—	0.162	-0.022	0.978
Core-7 minus PIV	1820	453	—	0.017 (0.003- 0.031)	—	—	—
Enhanced minus Core-7 (full cohort)	1820	453	—	0.010 (0.003- 0.017)	—	—	—

PIV, pan-immune-inflammation value. Enhanced = Core-7 plus fibrinogen. Performance estimates are based on repeated internal cross-validation.



227

228 Figure 2. Receiver operating characteristic curves from mean 5-fold-by-10-repeat held-out
 229 predictions; AUC labels are linked to the reported primary estimates.



230

231 Figure 3. Calibration by 10 quantile groups from the same mean 5-fold-by-10-repeat held-out
 232 predictions used for the reported calibration statistics.

233 **Clinical incremental value and fibrinogen enhancement**

234 The clinical baseline comprised age, sex, hypertension, and diabetes; smoking was
 235 unavailable. In the same fibrinogen-complete sample (N=1,705; AMI=426), AUCs were 0.546
 236 (95% CI 0.515-0.574) for Clinical, 0.717 (0.689-0.745) for Clinical+PIV, 0.730 (0.701-0.755) for
 237 Clinical+Core-7, and 0.741 (0.712-0.770) for Clinical+Enhanced. Hypertension was missing in
 238 79 patients and diabetes in 48; each was imputed using the training-fold median. Smoking was
 239 not added. The full paired comparison is in Additional file 2, Table S8.

Temporal and diagnostic sensitivity analyses

The exploratory deidentified date-axis allocation included development N=1001 (AMI=196), buffer N=271 (AMI=79), and later N=548 (AMI=178). In the later group, Core-7 AUC was 0.701, PIV 0.699, and Enhanced 0.719. The Core-7-minus-PIV difference was 0.002 (95% CI -0.021-0.025), indicating attenuation relative to the primary internal comparison. The high-specificity comparison contained 138 AMI and 288 strict controls; the 40 later-group AMI records not meeting the stricter text criteria were not adjudicated as false cases. Because date linkage to the index angiography hospitalization was incomplete, these results are not formal temporal validation. Exploratory decision-curve analysis is shown in Supplementary Figure S1.

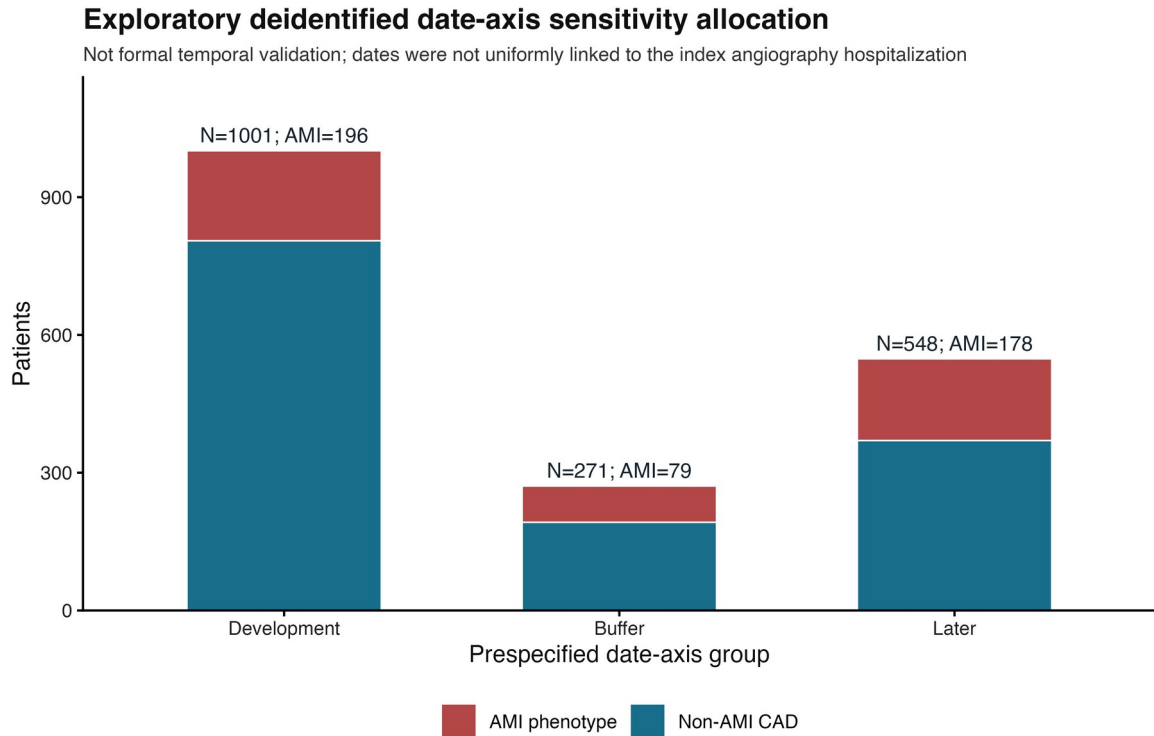


Figure 4. Exploratory deidentified date-axis sensitivity allocation; available dates were not uniformly linked to the index angiography hospitalization, so this is not formal temporal validation.

Table 3. Robustness analyses across diagnostic and temporal sensitivity cohorts

Analysis	N	AMI	Controls	PIV AUC (95% CI)	Core-7 AUC (95% CI)	Enhanced AUC (95% CI)
Exploratory deidentified date-axis later group	548	178	370	0.699 (0.651-0.745)	0.701 (0.650-0.748)	0.719 (0.673-0.764)
High-specificity phenotype comparison	426	138	288	0.749 (0.696-0.800)	0.770 (0.718-0.820)	0.787 (0.736-0.831)
Conservative +/-365-day date-axis sensitivity	361	Not reported	Not reported	0.698 (0.638-0.754)	0.684 (0.622-0.744)	0.699 (0.639-0.756)

Sensitivity analyses are summarized in Table 3 and Figure 5. Date-axis analyses are descriptive and should not be interpreted as temporal validation. The high-specificity comparison is based on discharge diagnosis text and is not prospective adjudication.

Predictor stability

Across 300 development bootstrap resamples, selection frequencies were 100% for neutrophils, lymphocytes, monocytes, platelets, and RDW, 99% for MPV, and 95% for hemoglobin. The maximum absolute pairwise Spearman correlation was 0.4685 between neutrophils and monocytes; no strong/extreme pairwise correlation was observed. These results are supportive stability evidence and do not justify removing variables. Additional results are provided in Additional files 1 and 2.

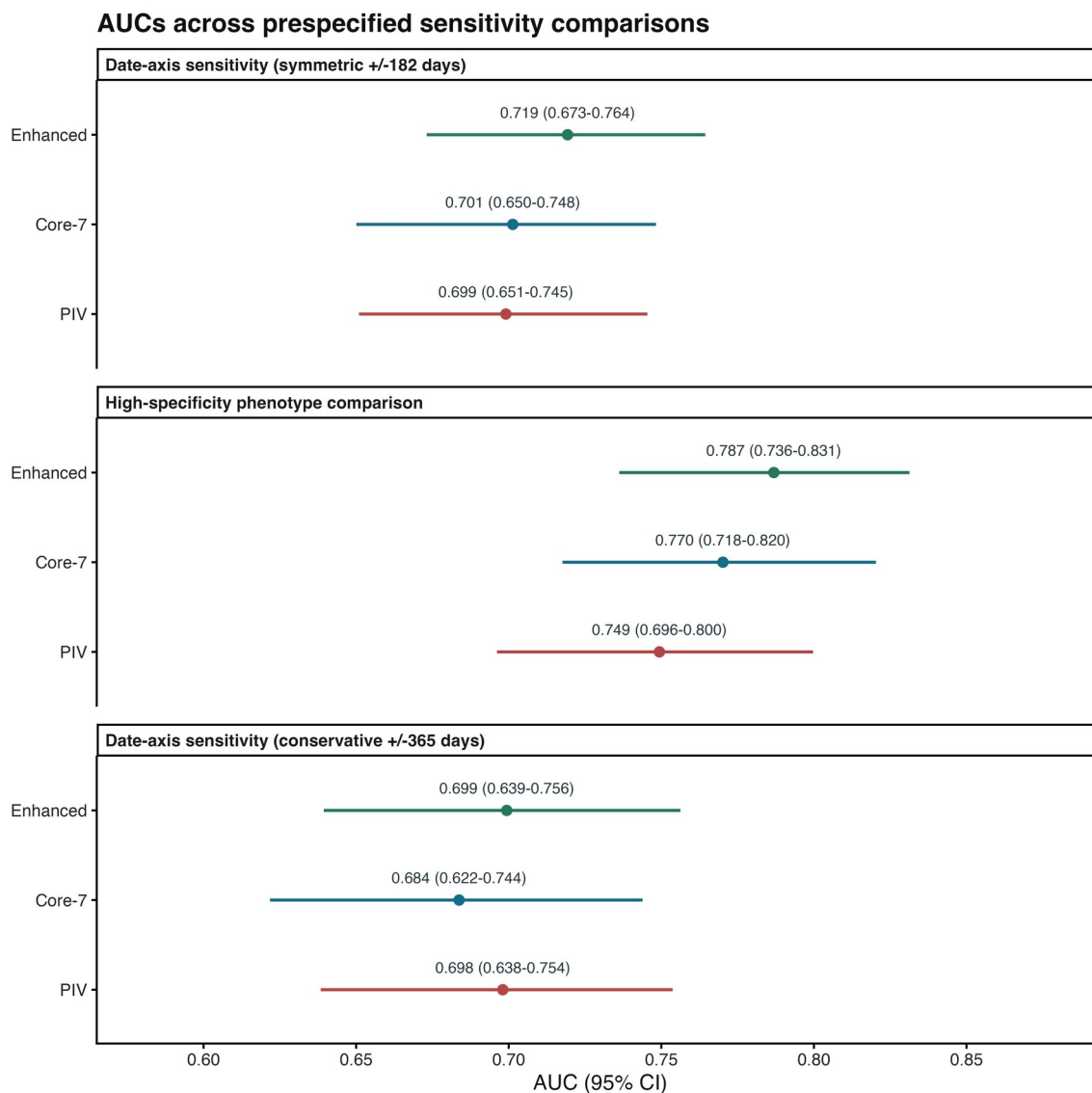


Figure 5. AUC estimates with 95% confidence intervals across prespecified date-axis and high-specificity phenotype sensitivity analyses.

Discussion

In this retrospective CAD cohort, Core-7 achieved an AUC of 0.725 (95% CI 0.697-0.753), with a modest paired increment over PIV (Δ AUC 0.017, 95% CI 0.003-0.031). This difference describes internal phenotype discrimination and does not establish diagnostic superiority.

271 The biological rationale for this multidimensional approach is consistent with the
272 pathophysiology of acute coronary events. Neutrophilia and lymphocyte depletion accompany
273 inflammatory and stress responses, while circulating monocytes participate in plaque
274 inflammation and the immune response after myocardial injury [1,3-12]. Platelet count and
275 MPV may reflect complementary aspects of platelet activation, consumption, and renewed
276 platelet production during acute coronary thrombosis [13,14]. RDW and hemoglobin can be
277 influenced by inflammation, erythropoietic stress, comorbidity, and oxygen-carrying reserve
278 [15,16]. Fibrinogen, as both an acute-phase protein and a coagulation substrate, adds
279 information from the thrombo-inflammatory axis that is not directly captured by the CBC [17].
280 These mechanisms provide a clinical rationale for considering the variables jointly, while the
281 individual regression coefficients should not be interpreted as causal effects or as evidence
282 that any single marker is protective or harmful.

283 The Core-7 advantage over PIV was attenuated in the exploratory deidentified date-axis
284 sensitivity analysis: AUCs were 0.701 and 0.699, respectively, with a paired Δ AUC of 0.002
285 (95% CI -0.021-0.025). The interval includes zero. Available dates could not be uniformly
286 linked to the index angiography hospitalization; the analysis is not formal temporal validation
287 and supports no transportability claim.

288 Fibrinogen was missing for 115/1,820 patients (6.32%), including 27/453 AMI (5.96%) and
289 88/1,367 non-AMI CAD patients (6.44%). In the same 1,705 fibrinogen-complete patients and
290 identical validation partitions, Enhanced AUC was 0.737 versus 0.727 for Core-7 (paired
291 Δ AUC 0.010, 95% CI 0.002-0.017), a small increment.

292 The present study was not designed to evaluate a real-time diagnostic, triage, or treatment-
293 decision tool. Selected CBC timing relative to the index angiography hospitalization, initial AMI
294 diagnosis, and treatment was not uniformly established. A pooled troponin comparison was not
295 methodologically valid because assay-platform details, ULN harmonization, serial change, and

index timing were incomplete; the study therefore cannot establish incremental value beyond a standardized contemporary troponin pathway.

This study is retrospective and single-center. The source export did not include a complete screening register or query record, so consecutive enrollment could not be confirmed.

Discharge-diagnosis availability contributed to the source-row ranking, so selection was not outcome-blind at the level of diagnosis availability; the magnitude of any resulting selection effect cannot be evaluated without the upstream extraction logic. Record consolidation used a completeness-based patient-level rule, but the retained row could not be linked uniformly to an index angiography hospitalization; selected CBC measurements therefore cannot be assumed to be admission-first, pre-diagnostic, or pre-procedural. AMI phenotype was based on discharge-diagnosis text without independent cardiologist, ECG, imaging, or standardized biomarker adjudication for all patients. AUC intervals were bootstrapped from fixed mean out-of-fold predictions and do not capture full model-development uncertainty. The exploratory deidentified date-axis analysis is not formal temporal validation, and no independent external cohort was available. Troponin assays could not be harmonized to establish incremental value beyond a standardized contemporary diagnostic pathway. These limits motivate external evaluation using prospectively time-anchored laboratories, complete source lineage, and standardized clinical adjudication.

Conclusions

In this single-center retrospective CAD cohort, joint modeling of routine hematologic measurements provided modest additional information for AMI phenotype discrimination beyond PIV, while fibrinogen contributed a further small increment. These findings are internally evaluated but should not be interpreted as evidence of diagnostic or triage readiness. Independent cohorts with prospectively time-anchored laboratory measurements and standardized clinical adjudication are required.

List of abbreviations

AMI, acute myocardial infarction; AUC, area under the receiver operating characteristic curve; CAD, coronary artery disease; CBC, complete blood count; CI, confidence interval; CK-MB, creatine kinase-MB; HRR, hemoglobin-to-red cell distribution width ratio; MLR, monocyte-to-lymphocyte ratio; MPV, mean platelet volume; NLR, neutrophil-to-lymphocyte ratio; NSTEMI, non-ST-segment elevation myocardial infarction; PCI, percutaneous coronary intervention; PDW, platelet distribution width; PIV, pan-immune-inflammation value; PLR, platelet-to-lymphocyte ratio; RDW, red cell distribution width; ROC, receiver operating characteristic; SII, systemic immune-inflammation index; SIRI, systemic inflammation response index; STEMI, ST-segment elevation myocardial infarction.

Declarations

Clinical trial number

Clinical trial number: not applicable.

Ethics approval

The study was approved by the Ethics Committee of Xinjiang Medical University (approval no. K202602-10). Given the retrospective nature of the study, the requirement for informed consent was waived by the Ethics Committee. All methods were performed in accordance with relevant guidelines and regulations.

Consent to participate

Informed consent was waived by the Ethics Committee.

Consent for publication

Consent for publication: not applicable.

Availability of data and materials

The clinical dataset is not publicly available because of institutional, ethical, and participant-privacy restrictions. Deidentified data may be available from the corresponding authors on reasonable request and subject to applicable institutional and ethics approval. Non-identifying analysis and figure-generation code is available from the corresponding authors on reasonable request.

Competing interests

The authors declare that they have no competing interests.

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Authors' contributions

JA: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing - original draft, Writing - review & editing, Visualization, Project administration. MP: Methodology, Validation, Investigation, Data curation, Writing - original draft, Writing - review & editing, Visualization. QX: Investigation, Resources, Data curation, Validation, Writing - review & editing. QP: Investigation, Resources, Data curation, Validation, Writing - review & editing. WJ: Investigation, Resources, Data curation, Validation, Writing - review & editing. BF: Investigation, Resources, Data curation, Validation, Writing - review & editing. FL: Methodology, Formal analysis, Validation, Writing - review & editing. LZ: Resources,

367 Validation, Writing - review & editing, Supervision. XL: Conceptualization, Resources, Writing -
368 review & editing, Supervision, Project administration, Funding acquisition. YY:
369 Conceptualization, Resources, Validation, Writing - review & editing, Supervision. All authors
370 read and approved the final manuscript.

371 Acknowledgements

372 Not applicable.

373 Additional files

374 Additional file 1. DOCX. Revised supplementary methods and Figures S1-S4.

375 Additional file 2. XLSX. Supplementary Tables S1-S13, including updated missingness,
376 conventional-index performance, calibration, clinical incremental comparisons, fibrinogen
377 complete-case results, high-specificity phenotype results, and exploratory date-axis sensitivity.

378 References

- 379 1. Hansson GK. Inflammation, atherosclerosis, and coronary artery disease. N Engl J Med.
380 2005;352:1685-1695. doi:10.1056/NEJMra043430.
- 381 2. Thygesen K, Alpert JS, Jaffe AS, et al. Fourth universal definition of myocardial infarction
382 (2018). Eur Heart J. 2019;40:237-269. doi:10.1093/eurheartj/ehy462.
- 383 3. Dziedzic EA, Gasior JS, Tuzimek L, et al. Investigation of the Associations of Novel
384 Inflammatory Biomarkers--Systemic Inflammatory Index and Systemic Inflammatory
385 Response Index--With the Severity of Coronary Artery Disease and Acute Coronary
386 Syndrome Occurrence. Int J Mol Sci. 2022;23:9553. doi:10.3390/ijms23179553.
- 387 4. Wei X, et al. Association of systemic immune inflammation index and system inflammation
388 response index with clinical risk of acute myocardial infarction. Front Cardiovasc Med.
389 2023;10:1248655. doi:10.3389/fcvm.2023.1248655.

- 390 5. Zhang C, et al. Systemic immune-inflammation index as a novel predictor of major adverse
391 cardiovascular events in patients undergoing percutaneous coronary intervention: a meta-
392 analysis of cohort studies. BMC Cardiovasc Disord. 2024;24:189. doi:10.1186/s12872-024-
393 03849-4.
- 394 6. Yang L, Guo J, Chen M, Wang Y, Li J, Zhang J. Pan-Immune-Inflammatory Value is
395 Superior to Other Inflammatory Indicators in Predicting Inpatient Major Adverse
396 Cardiovascular Events and Severe Coronary Artery Stenosis after Percutaneous Coronary
397 Intervention in STEMI Patients. Rev Cardiovasc Med. 2024;25(8):294.
398 doi:10.31083/j.rcm2508294. PMID:39228482.
- 399 7. Mangalesh S, et al. Development of a Novel Inflammatory Index to Predict Coronary Artery
400 Disease Severity in Patients With Acute Coronary Syndrome. Angiology. 2024;75:231-239.
401 doi:10.1177/00033197231151564.
- 402 8. Gao Y, et al. Prognosis impact of multiple novel lymphocyte-based inflammatory indices in
403 patients with initially diagnosed coronary artery disease. Immun Inflamm Dis.
404 2024;12:e1340. doi:10.1002/iid3.1340.
- 405 9. Zahorec R. Ratio of neutrophil to lymphocyte counts--rapid and simple parameter of
406 systemic inflammation and stress in critically ill. Bratisl Lek Listy. 2001;102:5-14.
407 PMID:11723675.
- 408 10. Li H, Zhou Y, Ma Y, Han S, Zhou L. The prognostic value of the platelet-to-lymphocyte
409 ratio in acute coronary syndrome: a systematic review and meta-analysis. Kardiol Pol.
410 2017;75(7):666-673. doi:10.5603/KP.a2017.0068. PMID:28394006.
- 411 11. Willim HA, Harianto JC, Cipta H. Platelet-to-Lymphocyte Ratio at Admission as a Predictor
412 of In-Hospital and Long-Term Outcomes in Patients With ST-Segment Elevation Myocardial
413 Infarction Undergoing Primary Percutaneous Coronary Intervention: A Systematic Review
414 and Meta-Analysis. Cardiol Res. 2021;12(2):109-116. doi:10.14740/cr1219.
415 PMID:33738014.

12. Ul Hussain H, Kumar KA, Zahid M, et al. Neutrophil to lymphocyte ratio as a prognostic marker for cardiovascular outcomes in patients with ST-segment elevation myocardial infarction after percutaneous coronary intervention: A systematic review and meta-analysis. *Medicine (Baltimore)*. 2024;103(26):e38692. doi:10.1097/MD.00000000000038692. PMID:38941418.
13. Galimzhanov A, Tenekecioglu E, Rustamova F, Tun HN, Mamas MA. The Prognostic Utility of Mean Platelet Volume in Patients With Acute Coronary Syndrome: A Systematic Review With Meta-Analyses. *Angiology*. 2022;73(8):734-743. doi:10.1177/00033197211070908. PMID:35062842.
14. Wang QC, Chen Y, Guan CL, Lu WK. Mean platelet volume and cardiovascular outcomes in patients with coronary heart disease: A meta-analysis of prospective studies. *Medicine (Baltimore)*. 2026;105(4):e41818. doi:10.1097/MD.00000000000041818. PMID:41578535. PMCID:PMC12851692.
15. Li H, Xu Y. Association between red blood cell distribution width-to-albumin ratio and prognosis of patients with acute myocardial infarction. *BMC Cardiovasc Disord*. 2023;23:66. doi:10.1186/s12872-023-03094-1.
16. Abraham LL 4th, Ramos JDA, Cunanan EL, Tiongson MDA, Punzalan FER. Red Cell Distribution Width and Mortality in Patients With Acute Coronary Syndrome: A Meta-Analysis on Prognosis. *Cardiol Res*. 2018;9(3):144-152. doi:10.14740/cr732w. PMID:29904449.
17. Cui Z, Zhao G, Liu X. Blood fibrinogen level as a biomarker of adverse outcomes in patients with coronary artery disease: A systematic review and meta-analysis. *Medicine (Baltimore)*. 2022;101(33):e30117. doi:10.1097/MD.00000000000030117. PMID:35984145. PMCID:PMC9387956.
18. Zou H, Hastie T. Regularization and variable selection via the elastic net. *J R Stat Soc Series B*. 2005;67:301-320. doi:10.1111/j.1467-9868.2005.00503.x.

- 442 19. Friedman J, Hastie T, Tibshirani R. Regularization Paths for Generalized Linear Models via
443 Coordinate Descent. *J Stat Softw.* 2010;33:1-22. doi:10.18637/jss.v033.i01.
- 444 20. Steyerberg EW, Vickers AJ, Cook NR, et al. Assessing the performance of prediction
445 models: a framework for traditional and novel measures. *Epidemiology.* 2010;21:128-138.
446 doi:10.1097/EDE.0b013e3181c30fb2.
- 447 21. van Calster B, McLernon DJ, van Smeden M, Wynants L, Steyerberg EW. Calibration: the
448 Achilles heel of predictive analytics. *BMC Med.* 2019;17:230. doi:10.1186/s12916-019-
449 1466-7.
- 450 22. Brier GW. Verification of forecasts expressed in terms of probability. *Mon Weather Rev.*
451 1950;78:1-3. doi:10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2.
- 452 23. Efron B, Tibshirani RJ. *An Introduction to the Bootstrap.* New York: Chapman & Hall; 1993.
453 doi:10.1007/978-1-4899-4541-9.
- 454 24. DeLong ER, DeLong DM, Clarke-Pearson DL. Comparing the areas under two or more
455 correlated receiver operating characteristic curves. *Biometrics.* 1988;44:837-845.
456 PMID:3203132.
- 457 25. Harrell FE Jr. *Regression Modeling Strategies: With Applications to Linear Models, Logistic
458 and Ordinal Regression, and Survival Analysis.* 2nd ed. Cham: Springer; 2015.
459 doi:10.1007/978-3-319-19425-7.
- 460 26. Riley RD, Snell KI, Ensor J, et al. Minimum sample size for developing a multivariable
461 prediction model: Part II--binary and time-to-event outcomes. *Stat Med.* 2019;38:1276-1296.
462 doi:10.1002/sim.7992.
- 463 27. Vickers AJ, Elkin EB. Decision curve analysis: a novel method for evaluating prediction
464 models. *Med Decis Making.* 2006;26:565-574. doi:10.1177/0272989X06295361.
- 465 28. von Elm E, Altman DG, Egger M, et al. The Strengthening the Reporting of Observational
466 Studies in Epidemiology (STROBE) statement. *PLoS Med.* 2007;4:e296.
467 doi:10.1371/journal.pmed.0040296.

- 468 29. Collins GS, et al. TRIPOD+AI statement: updated guidance for reporting clinical prediction
469 models that use regression or machine learning methods. *BMJ*. 2024;385:e078378.
470 doi:10.1136/bmj-2023-078378.
- 471 30. Moons KGM, et al. PROBAST+AI: an updated quality, risk of bias, and applicability
472 assessment tool for prediction models using regression or artificial intelligence methods.
473 *BMJ*. 2025;388:e082505. doi:10.1136/bmj-2024-082505.
- 474 31. R Core Team. R: A Language and Environment for Statistical Computing. Vienna, Austria:
475 R Foundation for Statistical Computing; 2026. doi:10.32614/R.manuals. [https://www.R-](https://www.R-project.org/)
476 [project.org/](https://www.R-project.org/).
- 477 32. Wickham H, Averick M, Bryan J, et al. Welcome to the tidyverse. *J Open Source Softw*.
478 2019;4:1686. doi:10.21105/joss.01686.